which, together with time, takes us to a total of eleven spacetime dimensions.

Moreover, Figure 12.7 illustrates a profound consequence of this new dimension. The structure of the Heterotic-E string changes as this dimension grows. It is stretched from a one-dimensional loop into a ribbon and then a deformed cylinder as we increase the size of the coupling constant! In other words, the Heterotic-E string is actually a two-dimensional membrane whose width (the vertical extent in Figure 12.7) is controlled by the size of the coupling constant. For over a decade, string theorists have always used perturbative methods that are firmly rooted in the assumption that the coupling constant is very small. As argued by Witten, this assumption has made the fundamental ingredients look and behave like one-dimensional strings even though they actually have a hidden, second spatial dimension. By relaxing the assumption that the coupling constant is very small and considering the physics of the Heterotic-E string when the coupling constant is large, the second dimension becomes manifest.

This realization does not invalidate any of the conclusions we have drawn in previous chapters, but it does force us to see them within a new framework. For instance, how does this all mesh with the one time and nine space dimensions required by string theory? Well, recall from Chapter 8 that this constraint arises from counting the number of independent directions in which a string can vibrate, and requiring that this number be just right to ensure that quantum-mechanical probabilities have sensible values. The new dimension we have just uncovered is not one in which a Heterotic-E string can vibrate, since it is a dimension that is locked within the structure of the “strings” themselves. Put another way, the perturbative framework that physicists used in deriving the requirement of a ten-dimensional spacetime assumed from the outset that the Heterotic-E coupling constant is small. Although it was not recognized until much later, this implicitly enforced two mutually consistent approximations: that the width of the membrane in Figure 12.7 is small, making it look like a string, and that the eleventh dimension is so small that it is beyond the sensitivity of the perturbative equations. Within this approximation scheme, we are led to envision a ten-dimensional universe filled with one-dimensional strings. Now we see that this is but an approximation